Programming and Mathematics Study Guide

A one-page guide with short programs, a list of key terms and two small tables.

Short Programs

Find a root by repeated halving.

```
def bisect(f, lo, hi, tol=1e-9):
    # f(lo) and f(hi) must have opposite signs
    while hi - lo > tol:
        mid = (lo + hi) / 2
        if f(lo) * f(mid) <= 0:
            hi = mid
        else:
            lo = mid
    return (lo + hi) / 2
```

Estimate an area with trapezoids.

```
def trapezoid(f, a, b, n=100):
    h = (b - a) / n
    total = (f(a) + f(b)) / 2
    for k in range(1, n):
        total += f(a + k * h)
    return total * h
```

Find a square root with Newton's method.

```
def newton(f, df, x, steps=8):
    for _ in range(steps):
        x = x - f(x) / df(x)
    return x

print(newton(lambda t: t * t - 2, lambda t: 2 * t, 1.0))
```

Sum the entries of an array in C.

```
int sum_array(const int *a, int n) {
    int total = 0;
    for (int i = 0; i < n; i++) {
        total += a[i];
    }
    return total;
}
```

Swap two integers through pointers.

```
void swap(int *p, int *q) {
    int tmp = *p;
    *p = *q;
    *q = tmp;
}
```

Summarise a column of numbers from the shell.

```
cut -d, -f2 grades.csv | sort -n | uniq -c | sort -rn | head
```

Count the lines of every data file.

```
for f in data/*.csv; do
    echo "$f has $(wc -l < "$f") lines"
done
```

Group quiz scores in a database query.

```
SELECT student, AVG(score) AS mean_score
FROM quiz
GROUP BY student
HAVING COUNT(*) >= 3
ORDER BY mean_score DESC;
```

Clamp a value to an interval in JavaScript.

```
function clamp(x, lo, hi) {
  return Math.min(hi, Math.max(lo, x));
}

console.log(clamp(12, 0, 10));
```

Check whether a word reads the same backwards.

```
def is_palindrome(s):
    cleaned = [ch.lower() for ch in s if ch.isalnum()]
    return cleaned == cleaned[::-1]
```

Sort a short list by hand.

```
def bubble_sort(items):
    items = list(items)
    for i in range(len(items)):
        for j in range(len(items) - 1 - i):
            if items[j] > items[j + 1]:
                items[j], items[j + 1] = items[j + 1], items[j]
    return items
```

Test a number for primality.

```
int is_prime(int n) {
    if (n < 2) return 0;
    for (int d = 2; d * d <= n; d++)
        if (n % d == 0) return 0;
    return 1;
}
```

Key Terms

Loop
A piece of code that repeats until a condition is met.

Recursion
A method in which a function is defined in terms of a smaller instance of itself.

Compiler
A program that translates source code into machine instructions.

Bit
The smallest unit of information, holding either zero or one.

Entropy
A measure of how many microscopic arrangements are consistent with a macroscopic state.

Torque
The turning effect of a force about an axis, equal to force times lever arm.

Resistance
The opposition of a material to the flow of electric current.

Refraction
The bending of a wave as it passes from one medium into another.

Enzyme
A protein that acts as a catalyst for a reaction in a living cell.

Tangent
A line that touches a curve at one point and has the same direction as the curve there.

Corollary
A result that follows almost immediately from a theorem that has just been proved.

Residual
The difference between an observed value and the value predicted by a model.

Outlier
A data point that lies unusually far from the rest of the values.

Dimensionless
Describing a quantity that has no unit, such as a ratio of two lengths.

Significant figures
The digits of a measured number that carry information about its precision.

Run Times of the Examples

Measured run times of four of the programs

Program	Input size	Steps	Time (ms)

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Bisection	1	30	0.02
Trapezoid rule	100	100	0.05
Sieve	10000	24000	1.4
Binary search	1000000	20	0.01

Reading and Schedule

Reading list. The chapter on loops and the short article on numerical error.

Schedule. Lecture on Tuesday, laboratory on Thursday and a quiz on Friday.